LAB-2(DAA)

CH.SC.U4CSE24220

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1.BUBBLE SORT

CODE:

```
#include <stdio.h>
void bubbleSort(int arr[],int n){
int i,j,temp;
for(i=0;i<n-1;i++){
for(j=0;j<n-i-1;j++){
if(arr[j]>arr[j+1]){
temp=arr[j];
arr[j]=arr[j+1];
arr[j+1]=temp;
}
}
}
}
int main(){
int n;
printf("Enter number of elements: ");
scanf("%d",&n);
int arr[n];
```

```
printf("Enter  elements");

for(int i=0;i<n;i++)

scanf("%d",&arr[i]);

bubbleSort(arr,n);

printf("elements after sorting");

for(int i=0;i<n;i++)

printf("%d\n",arr[i]);

return 0;

}
```

OUTPUT:

```
amma@amma17:~$ nano bubblesort.c
amma@amma17:~$ gcc bubblesort.c -o bubblesort
amma@amma17:~$ ./bubblesort
enter size5
enter elements1 2 6 4 5
elements after sorting
 1
 2
 4
 5
 6
```

2.INSERTION SORT

CODE:

```
#include <stdio.h>

void insertionSort(int arr[],int n){

int i,j,temp;

for(i=1;i<n;i++){

temp=arr[i];

j=i-1;
```

```
while(j>=0&&arr[j]>temp){
arr[j+1]=arr[j];
j--;
}
arr[j+1]=temp;
}
}
int main(){
int n;
printf("Enter elements: ");
scanf("%d",&n);
int arr[n];
printf("Enter %d elements: ",n);
for(int i=0;i<n;i++)
scanf("%d",&arr[i]);
insertionSort(arr,n);
printf("elements after insertion: ");
for(int i=0;i<n;i++)
printf("%d\n",arr[i]);
return 0;
}
```

OUTPUT:

```
amma@amma17:~$ nano insertionsort.c
amma@amma17:~$ gcc insertionsort.c -o insertionsort
amma@amma17:~$ ./insertionsort
enter size:6
enter elements:3
1
5
7
8
4
elements after insertion sort
1
3
4
5
7
8
```

3.selection sort

CODE:

```
#include <stdio.h>

void selectionSort(int arr[],int n){

int i,j,minIndex,temp;

for(i=0;i<n-1;i++){

minIndex=i;

for(j=i+1;j<n;j++)

if(arr[j]<arr[minIndex])

minIndex=j;

temp=arr[i];

arr[i]=arr[minIndex];

arr[minIndex]=temp;

}

}

int main(){
```

```
int n;
printf("Enter number of elements: ");
scanf("%d",&n);
int arr[n];
printf("Enter %d elements: ",n);
for(int i=0;i<n;i++)
scanf("%d",&arr[i]);
selectionSort(arr,n);
printf("Sorted array: ");
for(int i=0;i<n;i++)
printf("%d\n",arr[i]);
return 0;
}
```

OUTPUT:

```
amma@amma37:~$ nano selectionsort.c
amma@amma37:~$ gcc selectionsort.c -o selectionsort
amma@amma37:~$ ./selectionsort
Enter number of elements: 5
Enter 5 elements: 3
8
1
7
4
Sorted array: 1
 3
 4
 7
 8
```

4.HEAP SORT

CODE:

```
#include<stdio.h>

#define size 5

void heapify(int arr[], int n, int i){

int largest=i;

int left=2*i+1;

int right=2*i+2;

if(left<n && arr[left] > arr[largest]){

largest = left;

}

if(right<n && arr[right] > arr[largest]){

largest = right;

}

if(largest!=i){
```

```
int temp=arr[i];
arr[i] = arr[largest];
arr[largest]=temp;
heapify(arr,n,largest);
}
}

void heapsort(int arr[], int n) {
   for (int i = n/2 - 1; i >= 0; i--) {
      heapify(arr, n, i);
   }
   for (int i = n - 1; i > 0; i--) {
      int temp = arr[0];
      arr[0] = arr[i];
      arr[i] = temp;
      heapify(arr, i, 0);
   }
}
int main() {
   int arr[size];
   for (int i = 0; i < size; i++) {
      printf("enter element %d: ", i);
      scanf("%d", &arr[i]);
   }
   heapsort(arr, size);
   for (int i = 0; i < size; i++) {
```

```
    printf("%d\n", arr[i]);
  }
  return 0;
}
```

OUTPUT:

```
amma@amma11:~$ gcc heap_sort.c -o heap_sort
amma@amma11:~$ ./heap_sort
enter element 0: 9
enter element 1: 1
enter element 2: 8
enter element 3: 3
enter element 4: 6
1
3
6
8
9
```

5.BUCKET SORT

CODE:

```
#include <stdio.h>
void bucketSort(int arr[],int n){
int bucket[100]={0};
for(int i=0;i<n;i++)
```

```
bucket[arr[i]]++;
int k=0;
for(int i=0;i<100;i++){
while(bucket[i]>0){
arr[k++]=i;
bucket[i]--;
}
}
}
int main(){
int n;
printf("Enter number of elements: ");
scanf("%d",&n);
int arr[n];
printf("Enter %d elements: ",n);
for(int i=0;i<n;i++)
scanf("%d",&arr[i]);
bucketSort(arr,n);
printf("Sorted array: ");
for(int i=0;i<n;i++)
printf("%d\n",arr[i]);
return 0;
}
```

OUTPUT:

```
amma@amma37:~$ nano bucketsort.c
amma@amma37:~$ gcc bucketsort.c -o bucketsort
amma@amma37:~$ ./bucketsort
Enter number of elements: 5
Enter 5  values: 0.5
0.3
0.8
0.1
0.4
Sorted array: 0.100000 0.300000 0.400000 0.500000 0.800000
```